

Causal Inference Crash Course:

Part 10 — Conformal Inference

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Causal Inference Series

1. Foundations
2. Causal Models
3. Inference
4. Best Practices
5. Heterogeneous Treatment Effects
6. Panel Models
7. Regression Discontinuity
8. Surrogate Models
9. Arguable Validation
10. **Conformal Inference** ← **this deck**

Overview

- **Conformal inference** turns *any* model's point prediction into a **set** with a **finite-sample, distribution-free coverage guarantee** — it needs only **exchangeability**, not a distribution and not a correctly specified model.
- Why we care: honest uncertainty for the black-box predictors we actually use — and, the payoff, **valid causal inference with a single treated unit** (synthetic control), where classical standard errors don't apply.
- We cover:
 - the idea: **split conformal** and its coverage guarantee
 - the one assumption — **exchangeability** — and what breaks it
 - two fixes when it fails: **moving blocks** and **weighting**
 - the causal payoff: conformal inference for **synthetic controls**
- Assumes the earlier **Inference** and **Panel Models** decks.

The Idea

What we want

- Classical prediction intervals lean on a **model being right** — Wald needs a correct parametric form and large samples; Bayes needs a correct likelihood.
- **Conformal** asks for far less: a set $\mathcal{C}(x)$ with

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha$$

that holds in **finite samples**, for **any** data distribution, wrapped around **any** predictor.

- Running example: predict a new customer's next-quarter purchases and give an interval that really covers 90% of the time — whatever model produced the point prediction.

The one assumption: exchangeability

- The observations $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ are **exchangeable**: their joint distribution is unchanged by reordering them. (i.i.d. is the familiar special case.)
- That is the *whole* assumption — no distributional form, no correctly specified model. Everything downstream is a consequence of it.
- It is also exactly what **fails** for time series and panels — we will come back to that.

Conformity scores

- Pick a **nonconformity score** $s(x, y)$ — "how strange is this pair, to the model?" Larger means a worse fit.
- Regression default: the absolute residual of any fitted $\hat{f}$,

$$s(x, y) = |y - \hat{f}(x)|.$$

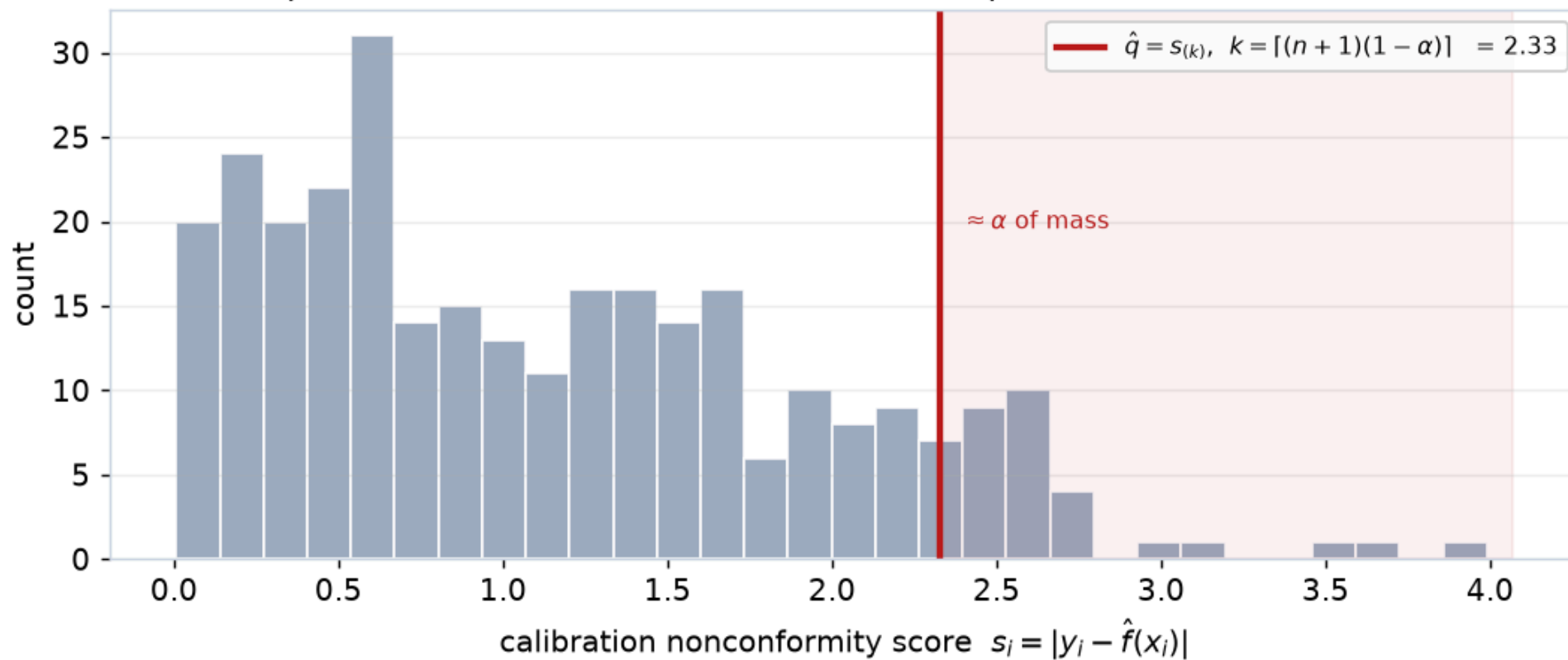
- The guarantee holds for **any** score and **any** model; a *good* score only makes the resulting set **tighter**, never invalid.

Split conformal — the recipe

For miscoverage level α , split the data into a **training** fold and a **calibration** fold of size n :

1. **Fit** $\hat{f}$ on the training fold only.
2. **Score** the calibration set: $s_i = s(X_i, Y_i)$, $i = 1, \dots, n$.
3. **Quantile**: $\hat{q} = s_{(k)}$, the k -th smallest score, with $k = \lceil (n + 1)(1 - \alpha) \rceil$.
4. **Predict**: $\mathcal{C}(x) = \{ y : s(x, y) \leq \hat{q} \}$.

Split conformal: threshold at the $(1-\alpha)$ quantile ($n=300$, $\alpha=0.1$)



The coverage guarantee

- **Theorem** (Vovk et al. 2005; Lei et al. 2018). If the data are **exchangeable**, split conformal satisfies

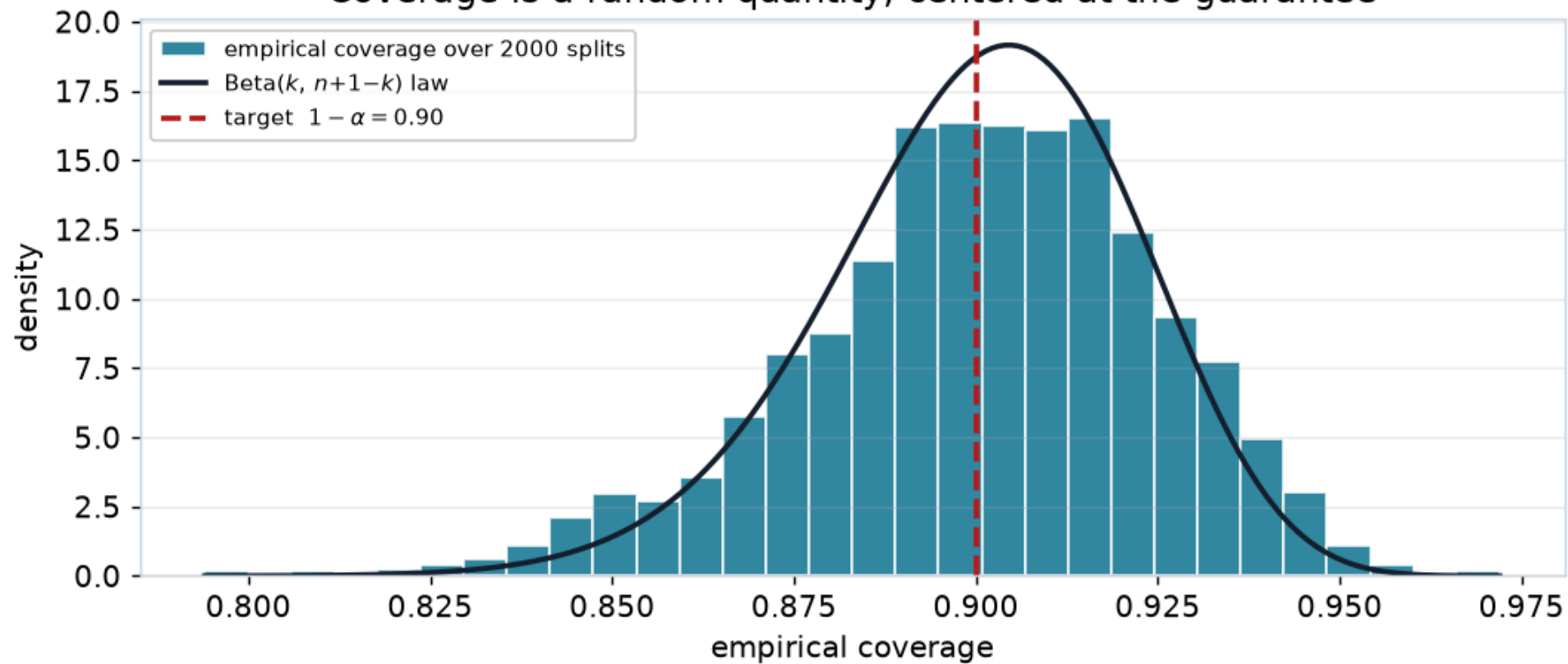
$$1 - \alpha \leq \mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \leq 1 - \alpha + \frac{1}{n+1}.$$

- **Finite-sample, distribution-free, model-free.** The upper bound needs only that scores have no ties.
- The model can be **arbitrarily wrong** — coverage still holds. A bad model simply yields an honestly **wide** set.

Why it works

- Insert the unseen test score as an $(n+1)$ -th point. By exchangeability its **rank is uniform** on $\{1, \dots, n+1\}$, so it exceeds the k -th calibration value only about α of the time — which is exactly the miscoverage event.
- So coverage is a **property of the procedure**, not of any one interval: over repeated draws it follows a $\text{Beta}(k, n+1-k)$ law centered at $1 - \alpha$.

Coverage is a random quantity, centered at the guarantee



The p-value view

- Equivalent framing (Shafer & Vovk 2008): conformal is **hypothesis testing turned into prediction**. For each candidate label y ,

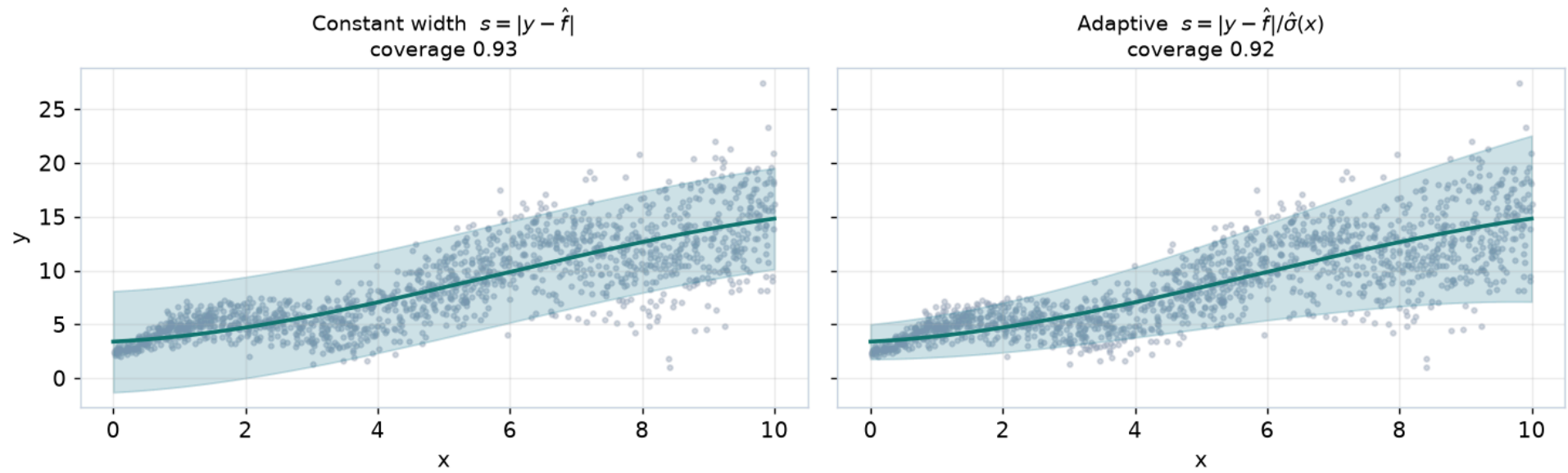
$$p(y) = \frac{1 + \#\{i : s_i \geq s(x, y)\}}{n + 1}, \quad \mathcal{C}(x) = \{y : p(y) > \alpha\}.$$

- Keep the labels you **cannot reject** at level α .
- This is the view we will use for synthetic control — where the "label" becomes a hypothesized treatment effect.

Adaptivity — and its limit

- The guarantee is **marginal**: it averages over draws. It is **not conditional** — some regions can be under-covered and others over-covered, and exact conditional coverage is impossible distribution-free.
- Fix with a smarter score, not a new guarantee: normalize by a local scale $s = |y - \hat{f}(x)| / \hat{\sigma}(x)$, or use **conformalized quantile regression** (Romano, Patterson & Candès 2019).

Same 90% marginal coverage, very different local behavior

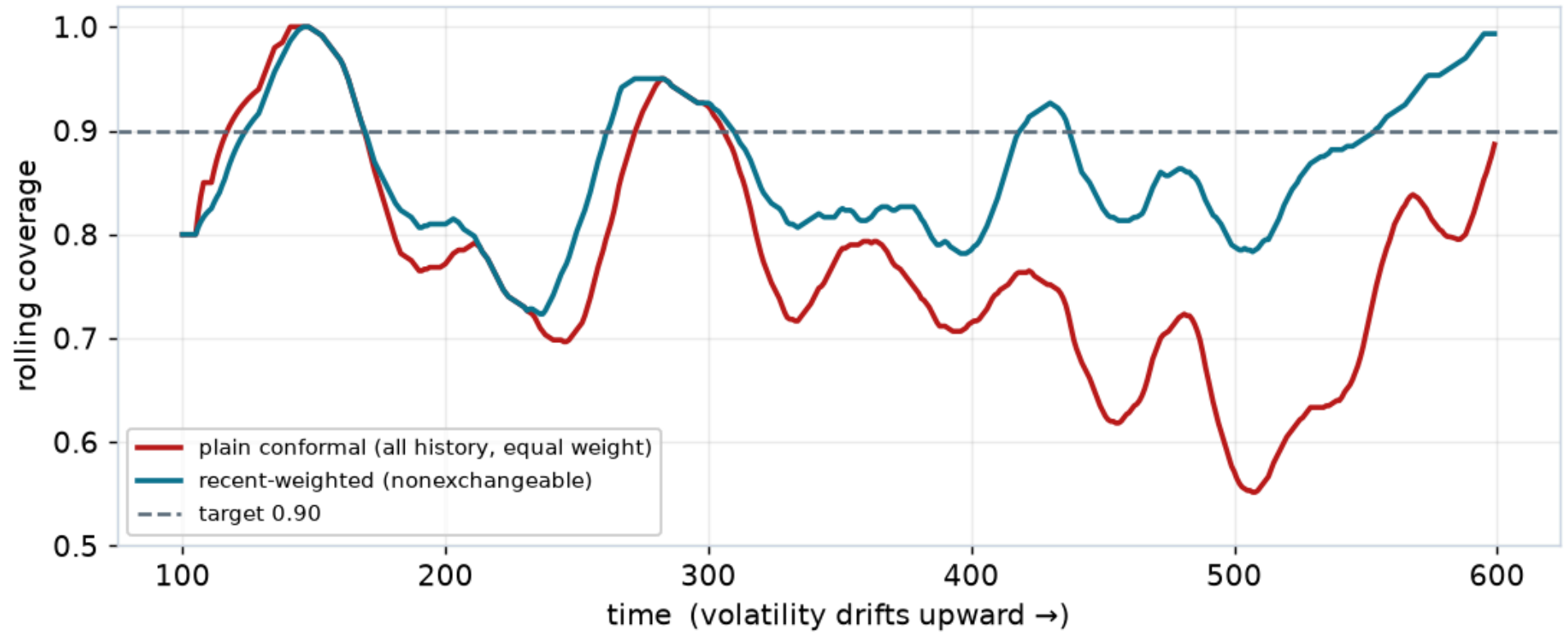


When Exchangeability Fails

Time and treatment break exchangeability

- Panels and time series are **serially dependent** and often **drift** — reorder them and the joint law changes, so exchangeability fails.
- Worse for causal work: the null holds only **before** treatment, and the post-period is exactly what we suspect differs.
- Plain conformal then **under-covers** — its interval is calibrated on a stale past that no longer resembles the present.

Under drift, exchangeability breaks and plain conformal under-covers



Fix A — permute blocks, not points

- Keep **contiguous blocks** of residuals intact and permute the block **positions** — the moving-block idea borrowed from the block bootstrap.
- Local dependence rides along *inside* each block; only its placement is randomized, so the null datasets keep the real time structure.
- **Exact** under exchangeability, **approximately valid** under stationarity and weak dependence — the block length must exceed the dependence range (Chernozhukov, Wüthrich & Zhu 2018, 2021).

Fix B — weight what belongs

- **Nonexchangeable conformal** (Barber, Candès, Ramdas & Tibshirani 2023): attach fixed weights w_i and take **weighted quantiles** of the scores — for time series, weight **recent** points more.
- The guarantee degrades gracefully:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha - \Delta,$$

where Δ is a weighted **distance from exchangeability**; it vanishes when the up-weighted points really do resemble the target.

- The weights must be **fixed a priori** — not learned from the scores.

Two relaxations of one idea

- **Blocks** — the *discrete* relaxation: glue dependent points together so a permutation can't shatter the local structure.
- **Weights** — the *continuous* relaxation: down-weight the points that don't belong with the target.
- Both **restore an approximate symmetry** the raw data lacks — just enough for the uniform-rank argument to (nearly) go through.

Conformal Inference for Synthetic Controls

The CWZ recast

- Chernozhukov, Wüthrich & Zhu (2021) glue two familiar problems together: causal inference = **counterfactual prediction** + **a test for a structural break**.
- Predict the treated unit's *untreated* path (synthetic control, DiD, factor models...), then ask whether the **post-treatment residuals break** from the pre-treatment ones.
- The pitch: **inference that doesn't care how good your synthetic control is** — validity comes from the permutation, not from a correct model.

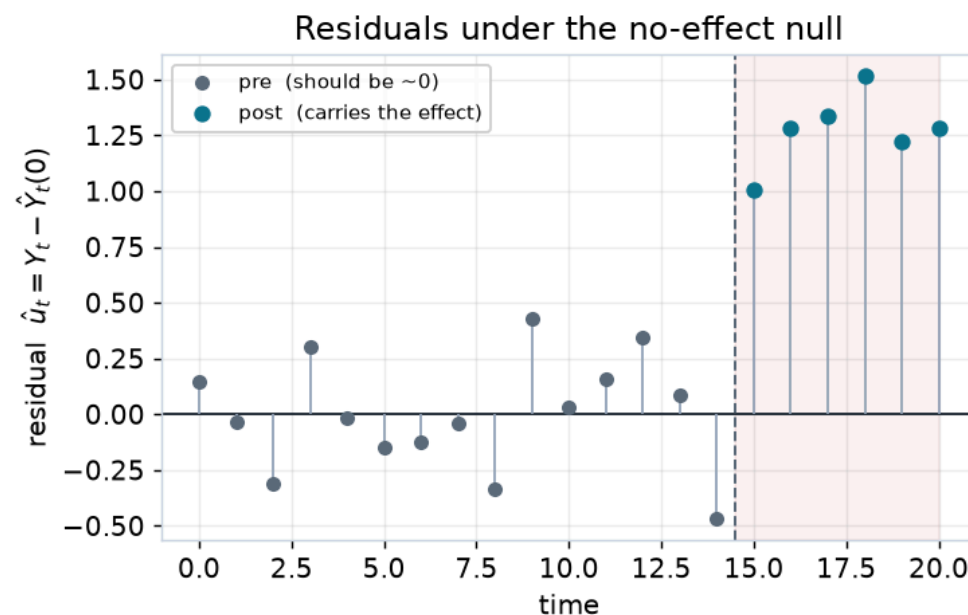
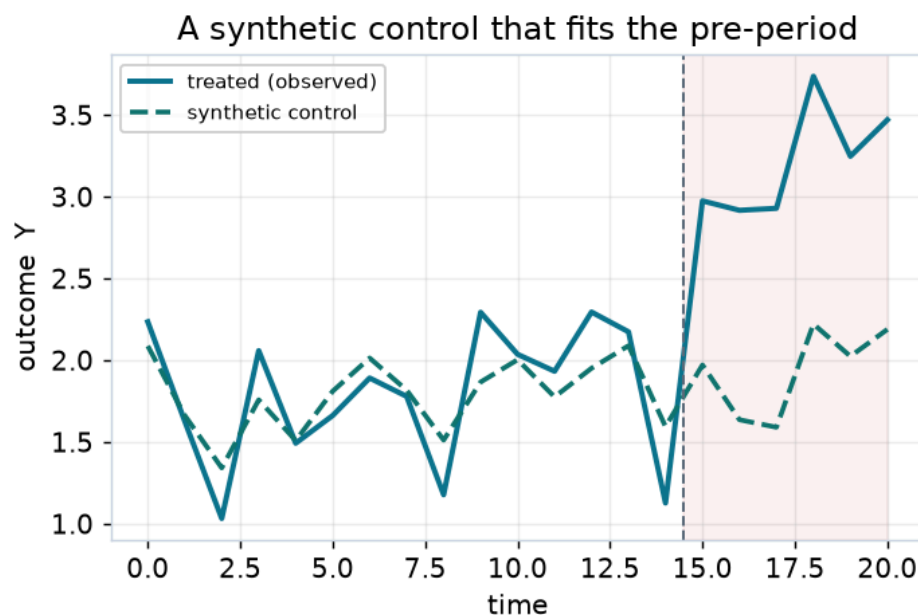
Setup — we predict the counterfactual

- One treated unit Y_{1t} , a donor pool Y_{0t} , pre-periods $t \leq T_0$ and post-periods $t > T_0$. Fit a **synthetic control** $\hat{Y}_{1t}(0)$ (from the Panel Models deck).
- The **gap** $\hat{u}_t = Y_{1t} - \hat{Y}_{1t}(0)$ is the residual. Under *no effect*, that residual process has **no break** at T_0 .
- The residuals over time are serially dependent — so this is exactly the non-exchangeable case, handled by **moving-block** permutations.

Step 1 — impose the null, read the residuals

- Test a sharp null on the post-period effect — here a constant effect
 $H_0 : \tau_t = \theta$. Subtract it from the **post outcomes only**, refit over the **whole timeline**, and take residuals for every t :

$$\tilde{Y}_{1t} = Y_{1t} - \theta \cdot \mathbf{1}\{t > T_0\}, \quad \hat{u}_t = \tilde{Y}_{1t} - \hat{Y}_{1t}(0)$$



Step 2 — a statistic and a moving-block permutation

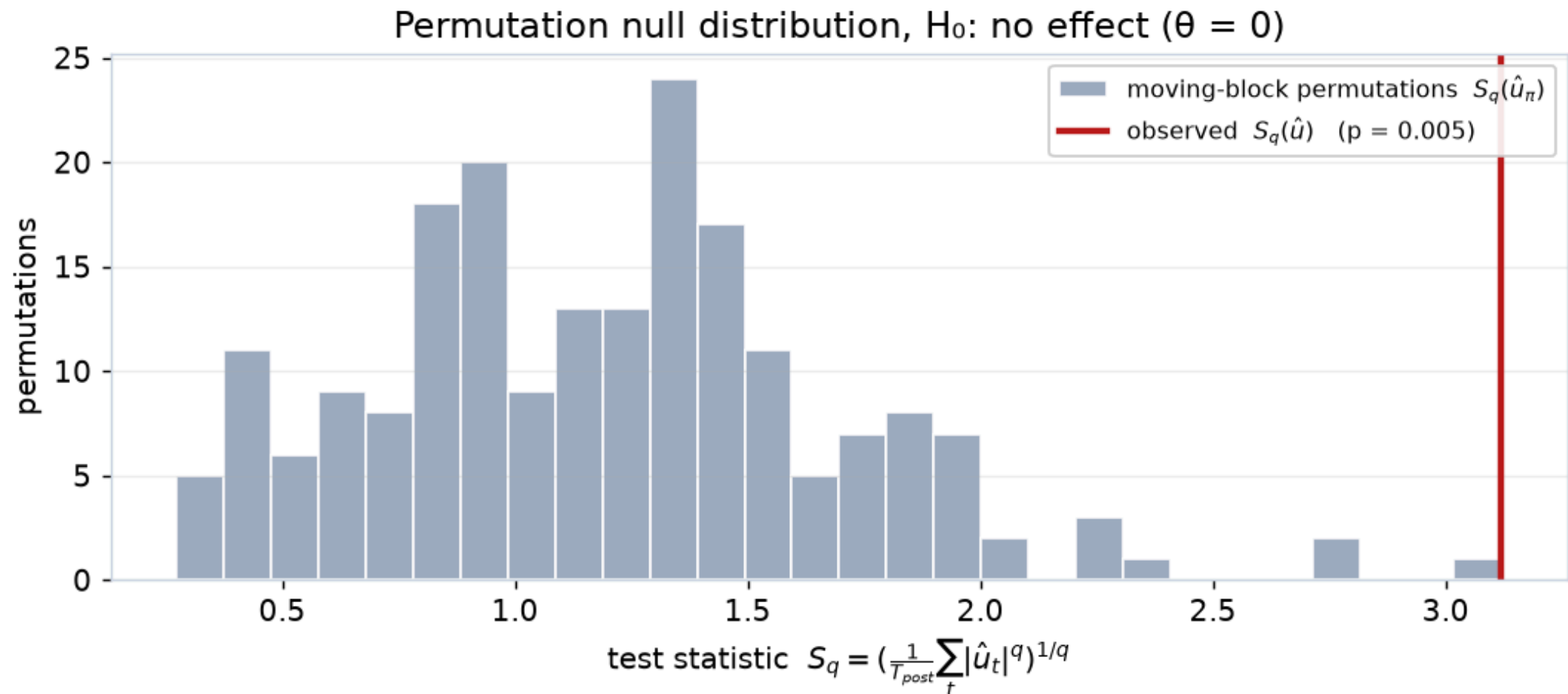
- Summarize the post-period residuals with a single number ($q = 1$ = mean absolute residual):

$$S_q(\hat{u}) = \left(\frac{1}{T_{post}} \sum_{t > T_0} |\hat{u}_t|^q \right)^{1/q}$$

- Compare S_q to its distribution under **moving-block permutations** of the residual sequence — the serial-dependence-robust scheme from the last section.
- `panelib`'s `conformal_inf` implements exactly this.

Step 3 — the permutation p-value

$$p = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \mathbf{1}\{S_q(\hat{u}_\pi) \geq S_q(\hat{u})\}, \quad \text{reject } H_0 \text{ if } p \leq \alpha.$$

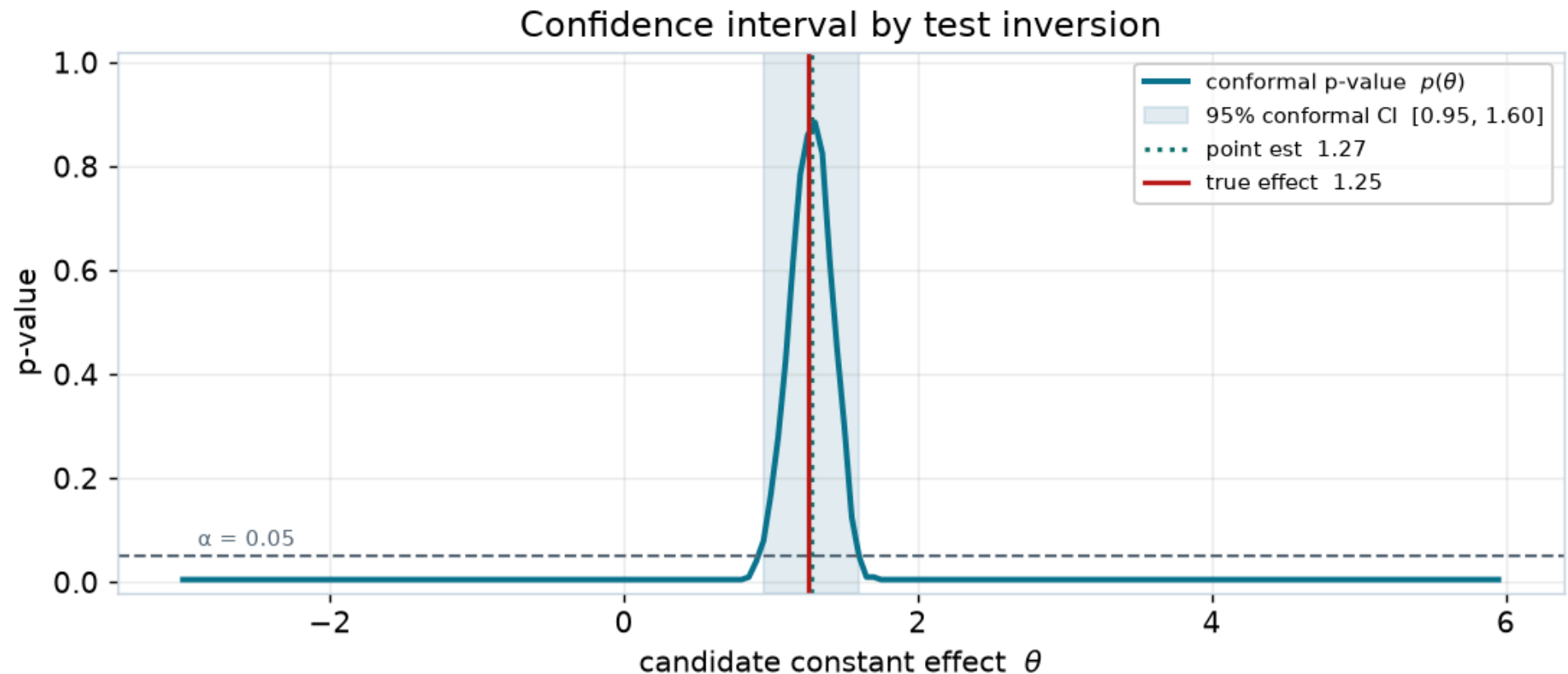


- **Careful:** this permutes residuals **across time** for one treated unit — not the Abadie placebo test, which permutes treatment **across units**.

Step 4 — confidence interval by test inversion

- Repeat the test over a grid of candidate effects θ and keep every value you **cannot reject**:

$$CI_{1-\alpha} = \{ \theta : p(\theta) > \alpha \}$$



- The point estimate sits where $p(\theta)$ peaks; the CI is the set of effects consistent with "no break."

Why this matters

- Works with **one treated unit and few pre-periods** — and is **exact** under exchangeability, **robust** to serial dependence and to a **misspecified** counterfactual (validity comes from residual symmetry, not a correct model).
- The fine print: under dependence the guarantee is approximate, and it needs the counterfactual's **error to be stable over time** — a model whose bias *trends* into the post-period breaks a symmetry no permutation can restore.
- Resolution matters too: with very few post-periods the permutation p-value is coarse — in the Panel deck's simulations (2 post-periods) coverage ran **80–85%**, not 95%.
- Contrast Abadie placebo inference: its p-value resolution is only $1/(J + 1)$ and it needs a large donor pool.
- This is the inference the **Panel Models** deck used to put DiD, SC, and SDID on the same footing.

Conclusion

- Conformal inference buys an **honest, finite-sample guarantee** from **one assumption you can actually reason about** — exchangeability — and degrades gracefully when it fails.
- Reach for it to **wrap black-box predictors**, and — the causal payoff — to do inference where classical SEs can't: **a single treated unit over time**.
- The price: coverage is **marginal**, not conditional; and under dependence you must **block** or **weight** to keep the guarantee.
- Start from the guarantee you can defend, then pick the score and the permutation that fit your data.

Appendix

Appendix: split vs. full conformal

- **Split (inductive)** conformal — the recipe here — spends part of the data on a calibration fold. Cheap (one fit), but the interval is random in that split and a bit wider.
- **Full** conformal refits for every candidate label — exact but expensive; **CV+ / jackknife+** (Barber et al. 2021) recover most of the efficiency without a held-out split.
- Same guarantee; the choice is **data efficiency vs. compute**.

Appendix: the coverage law

- Conditional on the calibration draw, the coverage of split conformal is a **random variable**:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C} \mid \text{calibration}) \sim \text{Beta}(k, n + 1 - k), \quad k = \lceil (n + 1)(1 - \alpha) \rceil.$$

- Its mean is $\geq 1 - \alpha$; the spread shrinks like $1/\sqrt{n}$ — the middle histogram made this concrete.
- The upper bound $1 - \alpha + \frac{1}{n+1}$ needs the scores to have **no ties** (continuous scores, or a random tie-break).

Appendix: another route — prediction intervals

- Conformal is not the only distribution-aware option for synthetic control.
- **Cattaneo, Feng & Titiunik (2021)** build **prediction intervals** that separate two error sources: the in-sample **weight-construction** error and the out-of-sample **stochastic** error, with finite-sample guarantees.
- Rule of thumb: **conformal** tests a sharp null by permutation and inverts to a CI; **prediction intervals** model the two error sources directly. Different assumptions, complementary diagnostics.

Appendix: conformal for treatment effects, beyond panels

- **Individual treatment effects** — Lei & Candès (2021): conformal intervals for $Y(1) - Y(0)$ under unconfoundedness, using *weighted* conformal to bridge the treated/control covariate shift. The natural companion to the HTE deck.
- **Covariate shift** — Tibshirani, Barber, Candès, Ramdas (2019): weighted conformal stays valid when the test point comes from a different covariate distribution with a known (or estimated) likelihood ratio.
- **Drift in production** — Gibbs & Candès (2021): *adaptive* conformal updates the level α_t online to keep realized coverage on target under distribution shift — the monitoring use case for deployed models.

Literature Review of Related Papers

- **Conformal prediction — foundations**

- Vovk, Gammerman, Shafer (2005). *Algorithmic Learning in a Random World*. Springer — the founding monograph. [link](#)
- Shafer & Vovk (2008). "A Tutorial on Conformal Prediction." *JMLR* 9, 371–421. [link](#)
- Lei, G'Sell, Rinaldo, Tibshirani, Wasserman (2018). "Distribution-Free Predictive Inference for Regression." *JASA* 113(523). [link](#)
- Angelopoulos & Bates (2023). "A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification." *Found. & Trends ML*. [link](#)

Literature Review (continued)

- **Adaptivity & beyond exchangeability**

- Romano, Patterson, Candès (2019). "Conformalized Quantile Regression." *NeurIPS*. [link](#)
- Barber, Candès, Ramdas, Tibshirani (2021). "Predictive Inference with the Jackknife+." *Ann. Statist.* 49(1). [link](#)
- Barber, Candès, Ramdas, Tibshirani (2023). "Conformal Prediction Beyond Exchangeability." *Ann. Statist.* 51(2), 816–845. [link](#)
- Tibshirani, Barber, Candès, Ramdas (2019). "Conformal Prediction Under Covariate Shift." *NeurIPS*. [link](#)
- Gibbs & Candès (2021). "Adaptive Conformal Inference Under Distribution Shift." *NeurIPS*. [link](#)

Literature Review (continued)

- **Conformal inference for causal panels / synthetic control**

- Chernozhukov, Wüthrich, Zhu (2018). "Exact and Robust Conformal Inference Methods for Predictive ML with Dependent Data." *COLT (PMLR 75)*. [link](#)
- Chernozhukov, Wüthrich, Zhu (2021). "An Exact and Robust Conformal Inference Method for Counterfactual and Synthetic Controls." *JASA* 116(536). [link](#)
- Cattaneo, Feng, Titiunik (2021). "Prediction Intervals for Synthetic Control Methods." *JASA* 116(536). [link](#)
- Firpo & Possebom (2018). "Synthetic Control Method: Inference, Sensitivity Analysis and Confidence Sets." *J. Causal Inference* 6(2). [link](#)
- Lei & Candès (2021). "Conformal Inference of Counterfactuals and Individual Treatment Effects." *JRSS-B* 83(5). [link](#)

